

Qn	Ans	Suggested Solution
1	C	Horizontal resultant force = $30.0 - 25.0 \sin 45^\circ = 12.3 \text{ N}$ Vertical resultant force = $10.0 - 25.0 \cos 45^\circ = 7.7 \text{ N}$ Resultant = $(12.3^2 + 7.7^2)^{1/2} = 14.5 \text{ m}$
2	D	Velocity is the gradient of the Displacement time graph.
3	D	Gradient of displacement time graph is velocity, hence option A, B and C are incorrect